

Medical Command Center

AI-Powered Insurance Claim Intake & Operations Platform

An end-to-end system for receiving, validating, extracting, reviewing and reporting on insurance claim documents — with a grounded AI assistant that never guesses.

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Executive Summary

Medical Command Center automates the intake and processing of insurance claim documents — from a raw faxed PDF to a reviewed, resolved claim record — using AI-assisted data extraction, automatic OCR, and a grounded conversational assistant, all wrapped in an operations dashboard built for claims teams.

Today, claims teams typically retype data from faxed or scanned documents by hand, cross-check figures manually, and have no easy way to ask "what changed on this claim?" without digging through paperwork. This platform removes that manual burden: a document is uploaded once, the system reads it, extracts every relevant field with a confidence score and a pointer back to exactly where it found each value, flags anything it isn't sure about, and hands the reviewer a clean, source-verified record to approve.

What makes this different

Grounded, not guessing

Every AI-extracted value and every AI chat answer is checked against the actual document text before it's shown. If the system can't verify something, it says so — it never invents patient, medical, or financial data.

Works with or without AI

A deterministic rule-based extraction engine runs underneath the AI layer, so the pipeline keeps working — with full transparency about which mode produced a given value — even if the AI provider is unavailable or rate-limited.

Built for review, not just automation

Low-confidence fields, conflicting values across pages, and manager corrections are all first-class concepts — corrections are protected from being silently overwritten by a later re-run.

Operational visibility

A live dashboard reports real claim volume, financial exposure, document quality, and an attention queue — filterable by date range, status, insurer and confidence — with CSV export.

Core Use Cases

1 Automated Document Intake

Staff upload a faxed or scanned PDF claim document directly — no manual re-keying of a paper or fax form is required to get the claim into the system.

2 Document Validation

Irrelevant or unrelated documents (e.g. a marketing flyer accidentally faxed to the same line) are detected and rejected before any data is fabricated for them.

3 AI-Assisted Field Extraction

Fifteen canonical claim fields — patient, policy, hospital, treatment dates, diagnosis and four distinct financial amounts — are extracted automatically with a per-field confidence score.

4 Automatic OCR for Scanned/Faxed Pages

When a page has no usable text layer (a true scanned fax), the system automatically runs OCR on that page — no separate "enable OCR" step for staff to remember.

5 Conflict & Source Verification

If a value appears differently on different pages of the same document, both candidates are surfaced rather than silently picking one. Every field links back to the exact page and text it came from.

6 Manager Review & Correction Workflow

Reviewers can correct any field, approve or reject a claim, and every action is written to an audit trail. A manager's correction is protected — a later re-run of extraction will not overwrite it.

7 Grounded AI Claim Assistant

A conversational assistant answers questions about the currently open claim — by voice or text — strictly from that claim's verified data, and asks for clarification rather than guessing when a question is ambiguous (e.g. "what's the amount?" when four different amount fields exist).

8 Operations Dashboard & Reporting

Claim volume, financial exposure, document/AI quality, insurer breakdowns, and a prioritized attention queue, filterable by date/status/insurer/confidence, with CSV export for further reporting.

System Architecture & Technology Stack

The system is split into a clean, layered backend and a modern single-page frontend, connected by a documented JSON API. Nothing about the AI/OCR layer is hard-wired into the UI or vice versa — each piece can be upgraded or swapped independently.

FRONTEND	
Framework	Angular 17 (standalone components, no legacy NgModules), TypeScript, RxJS
Routing	Full client-side routing — Dashboard, Fax Intake, Queued, Needs Review, Resolved, Failed, Deleted, each deep-linkable
Design system	Centralized theme tokens with a complete light and dark mode, responsive from desktop down to mobile (~390px)
Voice	Browser-native speech recognition for voice questions; real-time Web Audio analysis drives the AI avatar's lip-sync
BACKEND	
Framework	Python, FastAPI — layered into routers → services → repositories, so business logic never lives in a route handler
AI provider	Google Gemini (configurable model/key), abstracted behind a provider interface
AI fallback	Deterministic rule-based extraction engine — the app keeps functioning end-to-end with zero AI configured
Document parsing	PyMuPDF (native text + page rendering) and pdfplumber (table-aware text), with automatic Tesseract OCR for scanned pages
Voice synthesis	Offline, local text-to-speech engine (no cloud TTS credential required) generating real audio the frontend analyses for lip-sync
Data layer	Repository-pattern in-memory store today; designed so a persistent database can be substituted without changing service/route code
QUALITY & TESTING	
Automated tests	44 backend tests (pytest) covering extraction, OCR fallback, chat grounding, dashboard analytics, uploads, and text-to-speech
Build verification	Frontend verified in both development and production build configurations
Manual verification	Every feature in this report was exercised live in a real browser session before being marked complete

End-to-End Claim Workflow

- 1 Document received**

A staff member uploads a claim PDF (fax, scan, or digital document) from the Fax Intake screen.
- 2 Live processing progress**

The UI shows real, backend-reported progress through each pipeline stage — not a generic spinner.
- 3 Text extraction & automatic OCR**

Native text is extracted per page; any page without a usable text layer is automatically OCR'd, and tables are parsed separately so line items aren't lost.
- 4 Relevance validation**

The document is scored for how claim-relevant it is. Irrelevant documents are rejected outright; borderline ones require the reviewer to explicitly confirm before proceeding.
- 5 AI field extraction**

The fifteen canonical fields are extracted with a confidence score and an exact source-page reference for each.
- 6 Verification & conflict detection**

Each value is checked against the literal document text; if a field's value differs across pages, every distinct candidate is kept and flagged for review.
- 7 Manager review**

The reviewer corrects any field, sees which fields need attention, and approves the claim. Corrections are marked verified and protected from being overwritten.
- 8 Resolution & audit trail**

The claim moves to Resolved; every upload, correction, deletion, restoration and decision is recorded in an audit log.
- 9 Ask the AI Claim Manager**

At any point, the reviewer can ask the assistant a question about the open claim — by typing or by voice — and get a spoken, source-cited answer.
- 10 Operational reporting**

Every claim feeds the live dashboard: volume, financial exposure, document/AI quality, and a prioritized attention queue for the team lead.

How Hallucination Is Prevented

For an insurance claims workflow, an AI that occasionally invents a plausible-looking number is worse than no AI at all. This system enforces grounding in code, not just by asking the AI nicely in a prompt.

Code-level "found" check

The AI must explicitly report whether it actually found an answer. If it reports "not found," the system shows an honest "not available" message instead of trusting any text the model still produced.

Source-text verification

Whenever the AI cites a supporting snippet, the backend independently confirms that exact snippet really appears in the source document before it is ever shown to the user.

Deterministic fallback

Common factual questions (patient name, hospital, claim amount, etc.) are answered directly from the already-verified extracted fields — no AI call is even needed for these, so they can never fail due to an AI outage.

Honest failure states

A temporary AI/network problem is reported as exactly that — "the assistant is temporarily unavailable" — rather than being disguised as "this information doesn't exist," which would mislead a reviewer about the claim itself.

Manager authority is preserved

A value a human reviewer has corrected is marked manager-verified and is never silently overwritten by a later automatic re-extraction — automation defers to a human decision once one has been made.

Zero fabricated patient data

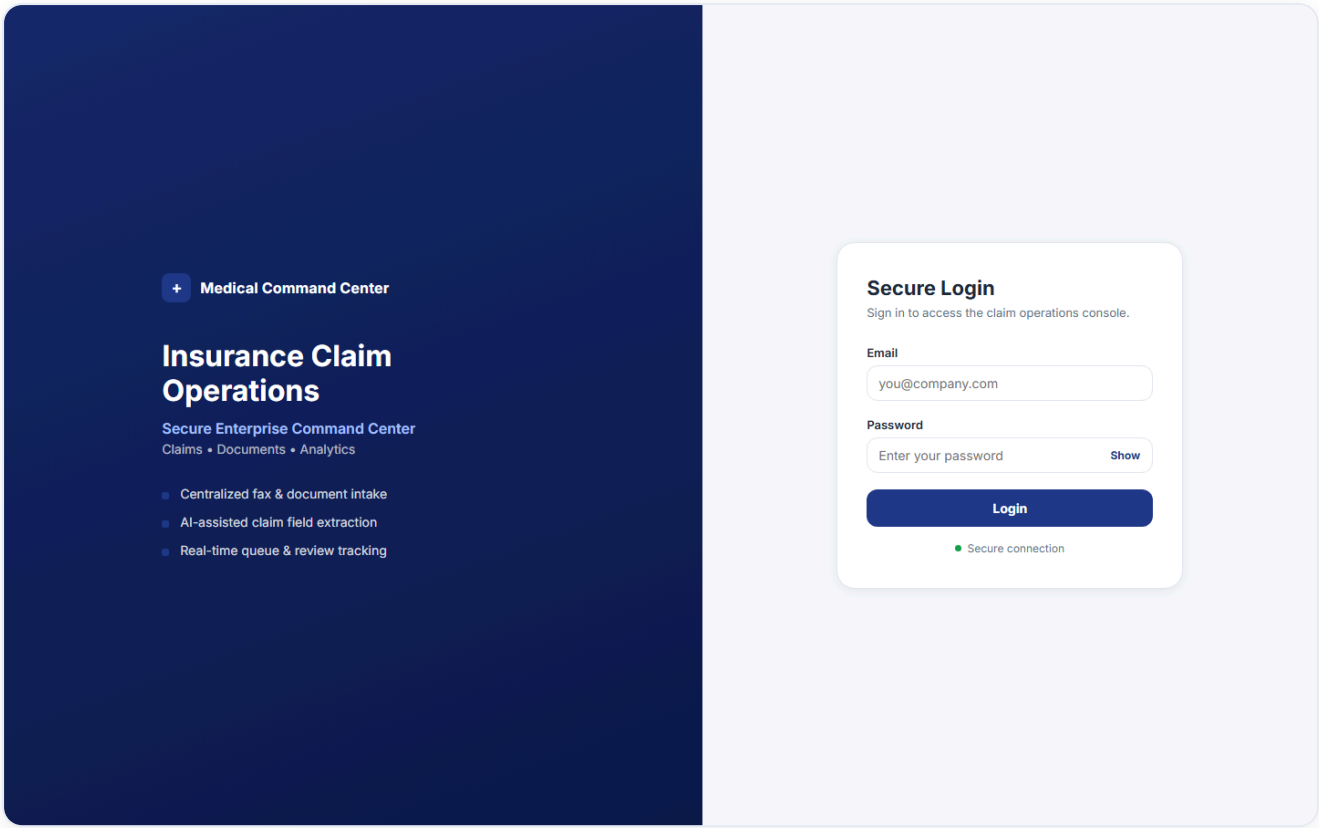
Zero fabricated financial figures

Every answer is source-cited

Ambiguous questions trigger clarification, not a guess

Secure Login

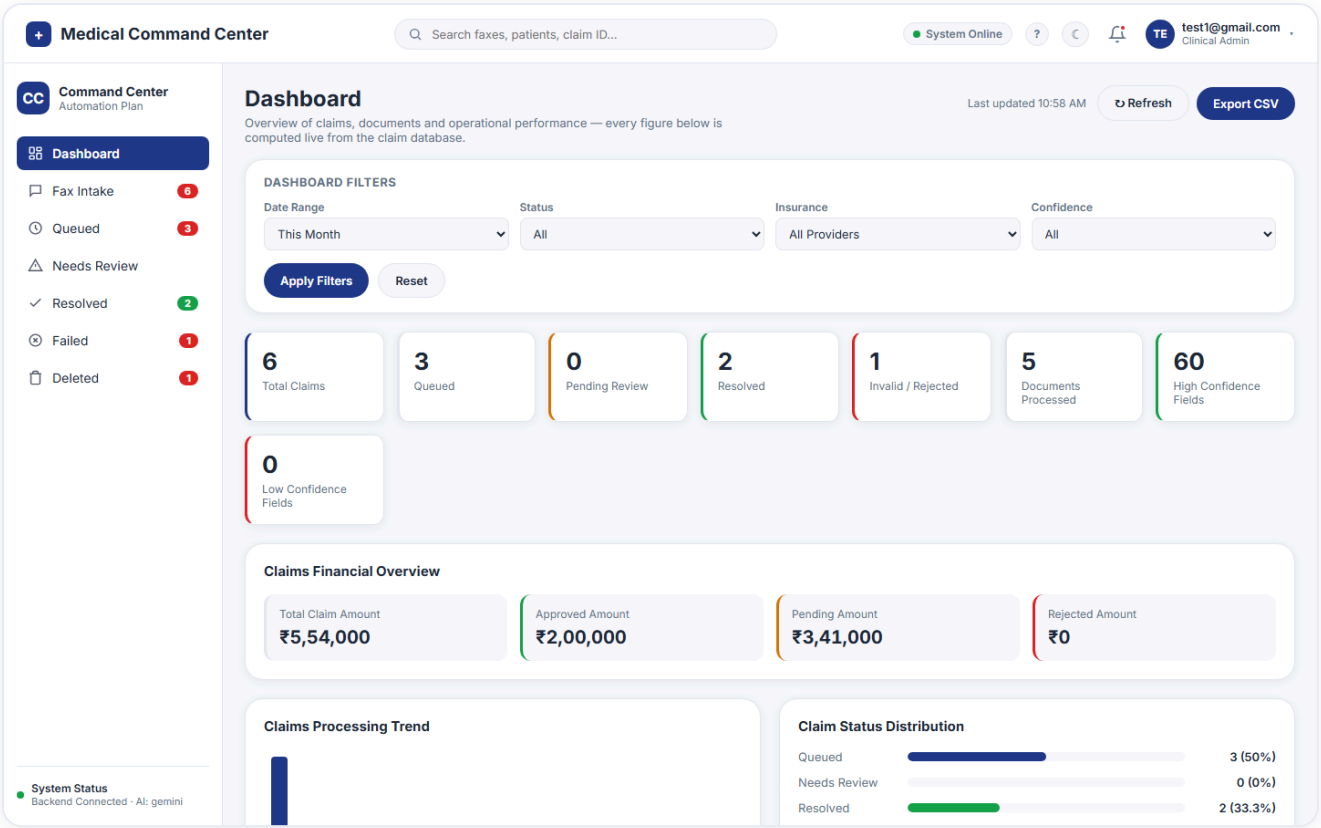
A professional, enterprise-styled sign-in screen gates every route in the app — the same visual identity carried through the entire product.



Secure sign-in screen, with the enterprise banking-style visual identity carried through the whole app.

Operations Dashboard

Every figure below is computed live from the claim database — total claims, financial exposure, document quality, and status distribution — filterable by date range, status, insurer, and confidence.



Live operations dashboard — every number is computed from real claim data, filterable by date range, status, insurer and confidence.

Claim Review

Each extracted field shows its own confidence score and a direct link back to the exact page and text it was sourced from, with the original document visible side by side for quick verification.

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Search faxes, patients, claim ID...

System Online

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test1@gmail.com
Clinical Admin

CC Command Center
Automation Plan

Dashboard

Fax Intake 6

Queued 3

Needs Review

Resolved 2

Failed 1

Deleted 1

System Status

Backend Connected - AI: gemini

Fax Intake 0 documents

+ Upload Fax (PDF)

PDFs only - scanned/faxed pages are OCR'd automatically

Upload a fax PDF above to run the AI extraction pipeline.

AI Auto-Fill Active — 89.3% Overall Confidence

All fields verified — ready for approval.
(rule-based extraction — configure AL_API_KEY for AI-powered analysis)

Re-run Extraction

Approve & Resolve Claim

Delete

Claim Information

100% Document Relevance

Patient Name 96% ✓

Priya Sharma

Page 1 — View Source

Date of Birth 95% ✓

22/07/1990

Page 1 — View Source

Member ID 88% ✓

M-77821

Page 1 — View Source

Insurance Company 90% ✓

Sunrise Health Insurance

Page 1 — View Source

Policy Number 90% ✓

SHI998877

Page 1 — View Source

Claim Number 92% ✓

CLM-2026-9081

Page 1 — View Source

Hospital Name 93% ✓

Lakeside Multispecialty Hospital

Page 1 — View Source

Doctor Name 88% ✓

Dr. Anil Mehta

Page 1 — View Source

Diagnosis 89% ✓

Page 1 — View Source

Admission Date 93% ✓

Page 1 — View Source

Source Document — professional_insurance_claim.pdf

SH
Sunrise Health Insurance
Insurance Claim Submission Form
Form Ref: MC-2026-0915
Secure Fax Transmission
Patient Information
Patient Name:
Priya Sharma
Date of Birth:
22/07/1990
Member ID:
M-77821
Insurance & Policy Details
Insurance Company:
Sunrise Health Insurance
Policy Number:
SHI998877
Claim Number:
CLM-2026-9081
Hospital & Treatment Details
Hospital Name:
Lakeside Multispecialty Hospital
Doctor Name:
Dr. Anil Mehta
Diagnosis:
Community acquired pneumonia

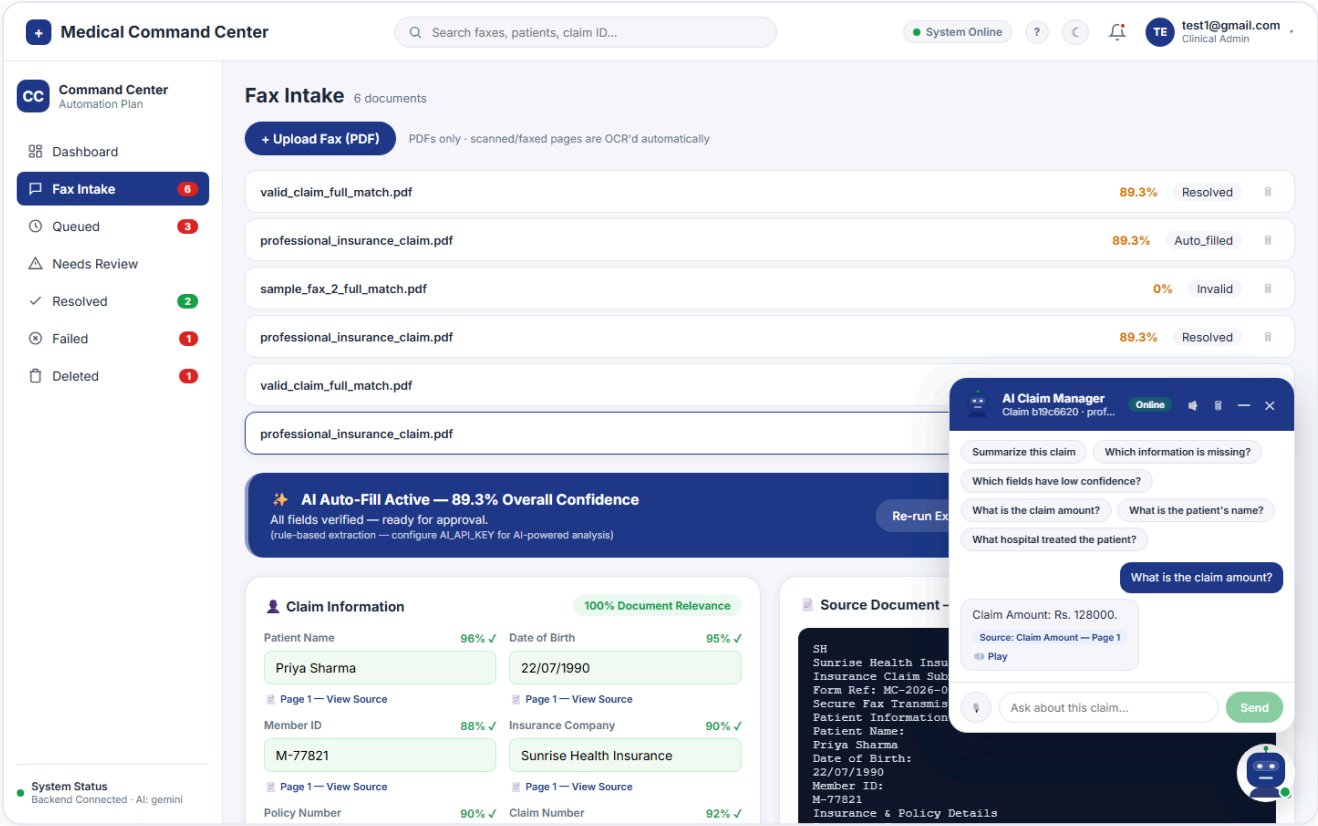
Claim review screen — extracted fields with per-field confidence, source links, and the original document side by side.

Medical Command Center — Confidential Demo Material

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The AI Claim Manager

A grounded conversational assistant answers questions about the open claim by text or voice, citing the exact source field for every answer — with a speaking avatar whose lip movement is driven by the real audio signal, not a looping animation.



The AI Claim Manager answering a grounded question about the open claim, with a source citation and voice playback.

Live Upload Progress

The stage label and percentage shown during upload reflect the actual backend pipeline in real time — uploading, text/OCR extraction, relevance checking, AI extraction, verification — not a simulated timer.

Medical Command Center

Search faxes, patients, claim ID...

System Online

?

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🔔

test1@gmail.com
Clinical Admin

CC Command Center
Automation Plan

Dashboard

Fax Intake 6

Queued 3

Needs Review

Resolved 2

Failed 1

Deleted 1

System Status
Backend Connected - AI: gemini

Fax Intake 6 documents

+ Upload Fax (PDF)

Uploading document... 2%

9DA0550AED338B7144E70F8353A05486E3FD7800

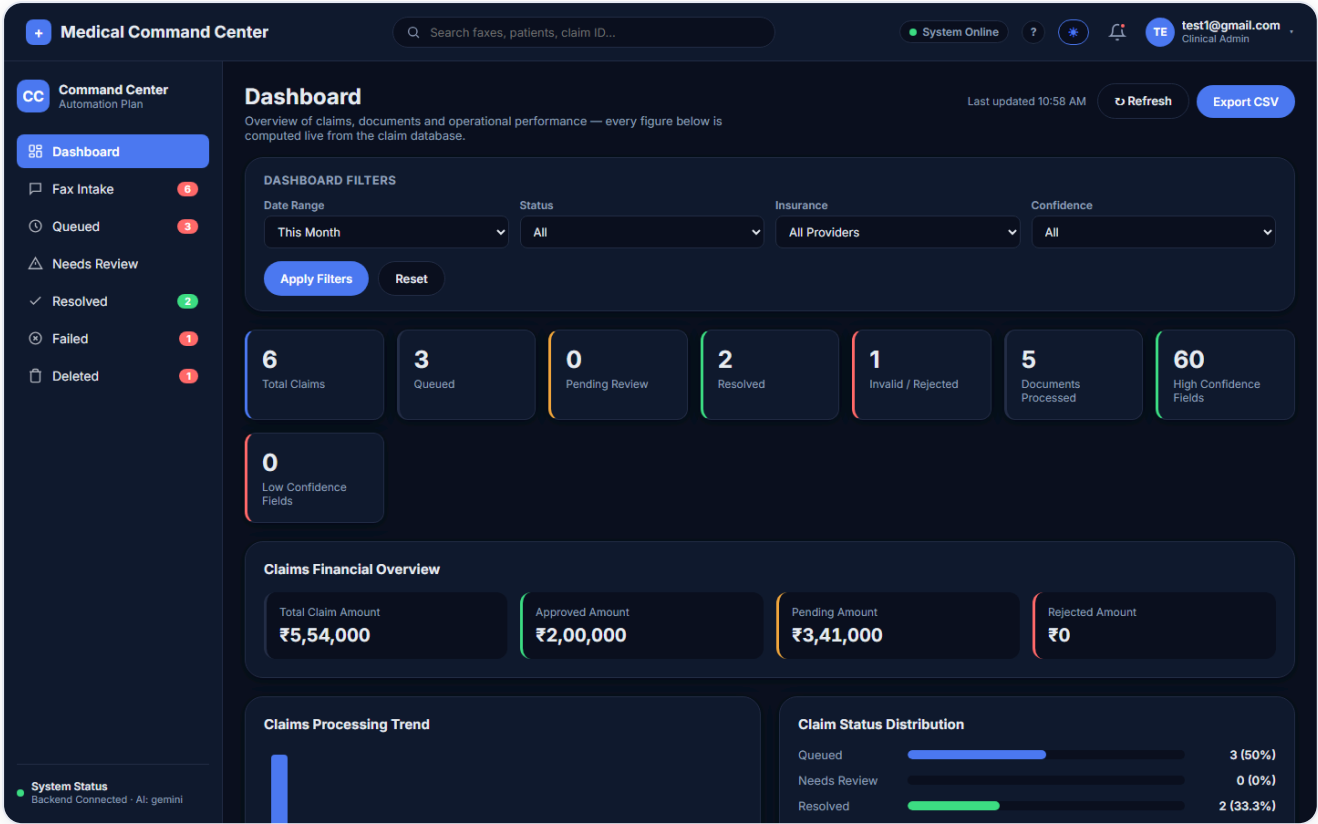
valid_claim_full_match.pdf	89.3%	Resolved	⋮
professional_insurance_claim.pdf	89.3%	Auto_filled	⋮
sample_fax_2_full_match.pdf	0%	Invalid	⋮
professional_insurance_claim.pdf	89.3%	Resolved	⋮
valid_claim_full_match.pdf	89.3%	Auto_filled	⋮
professional_insurance_claim.pdf	89.3%	Auto_filled	⋮

🤖

Real-time upload progress — the stage label and percentage reflect the actual backend pipeline, not a simulated timer.

Dark Mode

The full application supports a genuine dark theme with its own dedicated color palette — not simply an inverted filter — switchable from the header at any time.



The full application supports a genuine dark theme, not just an inverted color filter.

Current Demo Scope & Recommended Next Steps

In the interest of setting accurate expectations for a production rollout, this section is intentionally explicit about what today's build covers as a demo versus what a production deployment would add.

AREA	CURRENT DEMO BUILD	RECOMMENDED FOR PRODUCTION
Authentication	Single demo credential, for evaluation only	Real identity provider / SSO with role-based access control
Data storage	In-memory store (resets on server restart)	Persistent database — the codebase is already layered to add this without reworking business logic
Claim documents	One document per claim	Multi-document claim bundles
OCR engine	Automatic, requires the OCR engine installed on the host	Managed/containerized OCR service for guaranteed availability
AI provider	Google Gemini, with automatic offline fallback	Production-tier AI quota/SLA and provider redundancy

Ready to move forward?

The architecture underneath this demo was deliberately built in clean, swappable layers — the database, authentication, and AI provider can each be upgraded independently as the project moves from demo to production, with no rework of the extraction, review, or reporting logic already built and tested.